3D Vectors & the Cross Product

ANSWER KEY

Section 1: Dot Product and Orthogonality

1 Calculate the dot product $\mathbf{u} \cdot \mathbf{v}$ for $\mathbf{u} = [2, -3, 5]$ and $\mathbf{v} = [4, 1, -2]$.

a. -5 ✓

b. 5

c. 15

d. -1

$\mathbf{u} \cdot \mathbf{v} = (2)(4) + (-3)(1) + (5)(-2) = 8 - 3 - 10 = -5.$

2 For what value of k are the vectors $\mathbf{a} = [k, 3, -2]$ and $\mathbf{b} = [4, -2, 5]$ orthogonal (perpendicular)?

a. 4 ✓

b. 2

c. -4

d. 5

Orthogonal vectors satisfy $\mathbf{a} \cdot \mathbf{b} = 0 \Rightarrow 4k + 3(-2) + (-2)(5) = 0 \Rightarrow 4k - 6 - 10 = 0 \Rightarrow 4k = 16 \Rightarrow k = 4.$

3 Find the angle θ between $\mathbf{u} = [1, 1, 0]$ and $\mathbf{v} = [0, 1, 1]$.

a. $\frac{\pi}{3}$ **(**60°**)** ✓

b. $\frac{\pi}{4}$ (45°)

c. $\frac{\pi}{6}$ (30°)

d. $\frac{\pi}{2}$ (90°)

$\mathbf{u} \cdot \mathbf{v} = 1(0) + 1(1) + 0(1) = 1.\ |\mathbf{u}| = \sqrt{2}\,,\ |\mathbf{v}| = \sqrt{2}\,.\ \cos\theta = \frac{1}{\sqrt{2} \times \sqrt{2}} = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}.$

Section 2: The Vector Cross Product Formula

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4 Compute the cross product $\mathbf{u} \times \mathbf{v}$ where $\mathbf{u} = [1, 2, 3]$ and $\mathbf{v} = [4, 5, 6]$.

a. $[-3, 6, -3]$ ✓

b. $[3, -6, 3]$

c. $[-3, -6, -3]$

d. $[7, -2, -3]$

$\mathbf{u} \times \mathbf{v} = [2(6) - 3(5), -(1(6) - 3(4)), 1(5) - 2(4)] = [12 - 15, -(6 - 12), 5 - 8] = [-3, 6, -3]$.

5 What is the geometric relationship between the cross product vector $\mathbf{u} \times \mathbf{v}$ and the vectors $\mathbf{u}$ and $\mathbf{v}$?

a. **$\mathbf{u} \times \mathbf{v}$ is perpendicular to both $\mathbf{u}$ and $\mathbf{v}$** ✓

b. $\mathbf{u} \times \mathbf{v}$ is parallel to $\mathbf{u} + \mathbf{v}$

c. $\mathbf{u} \times \mathbf{v}$ bisects the angle between $\mathbf{u}$ and $\mathbf{v}$

d. $\mathbf{u} \times \mathbf{v}$ lies in the plane spanned by $\mathbf{u}$ and $\mathbf{v}$

By definition, the cross product produces a vector that is orthogonal to both input vectors, directed by the right-hand rule.

6 If $\mathbf{u} \times \mathbf{v} = \mathbf{0}$ for non-zero vectors $\mathbf{u}$ and $\mathbf{v}$, what does this indicate?

a. **The vectors are parallel (collinear)** ✓

b. The vectors are perpendicular

c. The vectors have unit magnitude

d. The vectors are identical

$|\mathbf{u} \times \mathbf{v}| = |\mathbf{u}||\mathbf{v}| \sin\theta = 0 \Rightarrow \sin\theta = 0 \Rightarrow \theta = 0$ or π, meaning the vectors are parallel.

Section 3: Area of Triangles and Parallelograms

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7 Find the area of the parallelogram spanned by vectors $\mathbf{a} = [2, 0, 1]$ and $\mathbf{b} = [1, 3, -2]$.

a. $\sqrt{59}$ ✓

b. $\sqrt{35}$

c. 14

d. $\frac{\sqrt{59}}{2}$

$\mathbf{a} \times \mathbf{b} = [0(-2) - 1(3), -(2(-2) - 1(1)), 2(3) - 0(1)] = [-3, 5, 6]$. Area $= |\mathbf{a} \times \mathbf{b}| = \sqrt{(-3)^2 + 5^2 + 6^2} = \sqrt{9 + 25 + 36} = \sqrt{70}$.

8 Find the area of the triangle with vertices $A(1, 0, 0), B(0, 2, 0), C(0, 0, 3)$.

a. $\frac{7}{2}$ ✓

b. 7

c. $\sqrt{14}$

d. 3

$\vec{AB} = [-1, 2, 0]$ and $\vec{AC} = [-1, 0, 3]$. $\vec{AB} \times \vec{AC} = [6, 3, 2]$. Magnitude $= \sqrt{36 + 9 + 4} = \sqrt{49} = 7$. Triangle area $= \frac{1}{2}(7) = 3.5$.

9 If two vectors have lengths $|\mathbf{u}| = 6$ and $|\mathbf{v}| = 5$, and the angle between them is 30°, what is $|\mathbf{u} \times \mathbf{v}|$?

a. 15 ✓

b. $15\sqrt{3}$

c. 30

d. $30\sqrt{3}$

$|\mathbf{u} \times \mathbf{v}| = |\mathbf{u}||\mathbf{v}|\sin(30^\circ) = (6)(5)\left(\frac{1}{2}\right) = 15$.

Section 4: Scalar Triple Product & Volume

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10 What geometric quantity is given by the absolute value of the scalar triple product $|\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|$?

a. **The volume of the parallelepiped formed by $\mathbf{a}, \mathbf{b}, \mathbf{c}$** ✓

b. The surface area of the prism formed by $\mathbf{a}, \mathbf{b}, \mathbf{c}$

c. The length of the main diagonal vector

d. The perimeter of the base

The scalar triple product $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$ represents the volume of the parallelepiped with adjacent edges $\mathbf{a}, \mathbf{b}, \mathbf{c}$.

11 Calculate the volume of the parallelepiped with edges $\mathbf{a} = [1, 0, 0]$, $\mathbf{b} = [0, 2, 0]$, $\mathbf{c} = [1, 1, 3]$.

a. 6 ✓

b. 3

c. 12

d. 2

Volume $= |\det[[1, 0, 0], [0, 2, 0], [1, 1, 3]]| = |1(2 \times 3 - 0)| = 6$.

12 If $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = 0$, what does this reveal about the three vectors?

a. **The three vectors are coplanar (lie in the same plane)** ✓

b. The three vectors are mutually perpendicular

c. At least one vector must be the zero vector

d. The vectors form an equilateral triangle

A zero scalar triple product implies zero enclosed volume, meaning the three vectors lie in a single common geometric plane (coplanar).

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